



Newman Smith DECA

Website — Structured Product Blueprint

newmansmithdeca.com | nshsdeca.com

MVP v1.0

1. Project Overview

Purpose	A centralized training platform for Newman Smith DECA members to practice roleplays, take practice tests, study Performance Indicators, and track progress — all in one place.
Primary URLs	newmansmithdeca.com / nshsdeca.com
Tech Stack	HTML, CSS, JavaScript — hosted on Vercel, database on Supabase
User Roles	Student, Officer (Pres, VP, etc), Advisor
MVP Scope	Core test-taking, roleplay submission & grading, PI study tools, user accounts
Future Scope	AI features, advanced analytics, district/state/ICDC personalization

2. User Roles & Authentication

2.1 Account System

Every user has a personal account with a username and password. User data is stored in Supabase and persists across sessions to enable progress tracking.

- Sign up with first name, last name, email, password, grade level
- Login / logout
- Profile page: display name, grade, event

2.2 Roles

Student (Default)

- Access all practice tests, study tools, and roleplay scenarios
- Submit roleplay videos for grading
- View own scores, PI strengths/weaknesses, and progress over time
- Earn points toward chapter leaderboard

Officer (Admin)

- All student capabilities
- Access submitted roleplays from any student for grading
- Leave timestamped video feedback (EdPuzzle-style)
- Submit rubric scores for student submissions
- View chapter-wide progress dashboards
- Manage announcements and chapter notices

President/Advisor

- All officer capabilities
 - Manage officer accounts and grant/revoke officer status
 - Export chapter progress data (CSV)
 - Full admin access to all site content and settings
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3. Practice Tests

3.1 Question Bank

Questions are scraped from deca.org/resources and decaplus.org. Each question stores metadata to enable smart filtering and PI tracking.

- Question text + 4 answer choices (A–D)
- Correct answer
- Rationale / explanation for correct and incorrect answers
- Associated PI code(s)
- Cluster, instructional area, level (district/state/ICDC)
- Source (deca.org, decaplus.org, etc.)

3.2 Test Modes

Full Practice Test

- Standard 100-question test format
- Timed or untimed option
- Graded at the end with full score breakdown

Custom Test

- User selects number of questions (e.g. 10, 25, 50, 100)
- Filter by cluster, instructional area, specific PIs, or level
- Choose grading mode: immediate feedback per question OR full test review at end

PI-Targeted Test

- Auto-generates a test based on user's weakest PIs
- Pulls questions specifically tagged to those PIs
- Future implementation: AI-generated questions for PIs with limited question bank coverage

3.3 During-Test Features

- Calculator (standard, accessible via button)
- Notepad / scratch pad — in-session notes visible during and after test
- Flag question for review
- Progress bar showing questions answered vs. remaining
- Stopwatch (optional, user-toggled)

3.4 Post-Test Review

- Full score: X/100, percentage, estimated scaled score
- Question-by-question breakdown with correct answer + rationale
- PI performance summary: which PIs were strong vs. weak this test
- Save notes taken during test to personal notebook
- Compare this attempt to past attempts on the same test

3.5 Progress Tracking

- Every question answered is logged to Supabase with the PI it belongs to
- User dashboard shows rolling PI accuracy across all tests taken
- Heatmap or bar chart of PI performance by cluster
- Test history log: date, test type, score, # questions
- Streak / consistency tracker (days studied)

4. Roleplay System

4.1 Scenario Library

Roleplay scenarios are sourced from deca.org/resources and decaplus.org where possible, and stored in the database with full metadata.

- Scenario title, situation description, student instructions, judge instructions
- Associated event, cluster, instructional area, PI(s)
- Year and competition level (district, state, ICDC)
- Prep time and presentation time
- Allowed materials
- Filter/search by event, cluster, PI, year, instructional area

4.2 Roleplay Submission

Students record and submit their roleplay attempts for grading by officers.

- Student selects a scenario from the library
- Uploads a video via Google Drive link embed
- Submission includes: scenario ID, event, attempt number, video URL
- Submission status: Submitted → Under Review → Reviewed

- Students can re-submit (multiple attempts tracked separately)

4.3 Grading Interface (Officer View)

Officers access an EdPuzzle-style video grading interface where they can leave timestamped feedback while watching the submission.

- Embedded Google Drive video player
- Timeline scrubber with comment markers visible at each timestamp
- Click any point on the timeline to add a timestamped text comment
- Comments tagged by PI or rubric category
- Graders submit final rubric scores after watching

Rubric Scoring

- PI scores (per PI evaluated in the event)
- 21st Century Skills score
- Total score out of rubric maximum
- Fill out rubric
- Free-text overall feedback field
- Auto-saves in progress; officer submits when complete

4.4 Student Feedback View

- Student sees submitted rubric scores and total
- Plays back their own video with officer comments overlaid at timestamps
- PI breakdown: which PIs scored well, which need improvement
- Notes section: officer's written feedback
- All past submissions and scores stored in history

4.5 PI Weakness Tracking from Roleplays

- Each graded rubric automatically feeds into student's PI profile
- Over time, student dashboard shows cumulative PI accuracy across both tests and roleplays
- Weak PIs surfaced as recommendations: 'Study these PIs before your next roleplay'
- Future: AI-generated custom roleplay scenario targeting weak PIs

5. Study Tools

5.1 Performance Indicators (PI) Browser

A full database of all DECA Performance Indicators, sourced from the official PI sheet. Sortable and filterable by any attribute.

- PI code, full indicator text, performance element, instructional area, cluster, tier/level
- Filter by: cluster, instructional area, performance element, level, pathway
- Search by keyword
- Click any PI to open a detail page with definition, examples, and study options

5.2 Flashcards

PI Flashcards

- Premade flashcard sets for every cluster and instructional area
- Card front: PI code + performance indicator text
- Card back: full definition, context, and example application
- User can write their own definition before flipping to compare
- AI-generated explanation for any PI on demand

Vocab Flashcards

- Premade vocab sets per cluster and per event type
- Cards include: term, definition, cluster tag, example usage in context
- User can create custom vocab cards
- Future: AI-generated vocab cards for any topic

5.3 Study Modes

Learn Mode

- Guided card-by-card review with spaced repetition logic
- Marks cards as 'still learning' or 'know it'
- Prioritizes cards marked 'still learning' in future sessions

Test Mode

- User types answer or selects from multiple choice
- Scored at end with detailed breakdown
- Wrong answers added to review queue

Match Mode

- Drag-and-drop matching game: PI/term on one side, definition on the other
- Timed or untimed
- Score based on accuracy and speed

5.4 Psychological Theories & Motivational Frameworks

A dedicated section covering the most common theories tested in DECA roleplays and exams.

- Theory name, full explanation, origin/context
- Real-world example scenario for each theory
- MCQ drill: given an example situation, identify which theory applies
- Covers: Maslow, Herzberg, McGregor (Theory X/Y), Vroom, Equity Theory, etc.
- Logical fallacies: definitions + examples + MCQ identification
- Psychological sales/persuasion principles used in marketing events

5.5 Visual Reference Library

A library of visual aids commonly referenced in DECA roleplays and exams, organized by cluster.

- Charts, graphs, matrices (BCG, SWOT, Ansoff, etc.)
- Example visuals from past real DECA roleplay recordings
- Each visual labeled with name, use case, cluster relevance
- Filter by cluster or visual type

5.6 Personal Notes

- A full-featured personal notes section within the site
 - Multiple tabs / notebooks (e.g. one per event or cluster)
 - Rich text editor: bold, italic, lists, headers
 - Auto-save to Supabase
 - Notes from MCQ tests can be saved directly here
 - Search across all notes
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6. Gamification & Leaderboard

6.1 Points System

Action	Points
Complete a practice test (100q)	10 pts
Score 80%+ on a test	+5 bonus pts
Submit a roleplay for grading	15 pts
Study flashcards (per 20-card session)	5 pts/day cap
7-day study streak	+15 bonus pts

6.2 Leaderboard

- Chapter-wide leaderboard ranking top 10 members by total points
- Top 10 Weekly and all-time views
- Shows: display name, grade level, total points, # tests, # roleplays
- Students can opt out of public leaderboard display
- Officers can pin/feature top performers in announcements

6.3 Achievements / Badges (Future)

- 'First Roleplay Submitted', 'Test Streak 7 Days', '100 Questions Answered', etc.
 - Displayed on profile page
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7. Site Pages & Navigation

7.1 Page Map

- / — Home / Landing (chapter info, quick links, announcements)
- /login — Login / Sign up
- /dashboard — Personal dashboard (scores, weak PIs, recent activity)
- /tests — Test hub (browse, filter, start test)

- /tests/:id — Active test session
- /tests/:id/review — Post-test review
- /roleplays — Scenario library (filter by event, cluster, PI, year)
- /roleplays/:id — Scenario detail + submission form
- /submissions/:id — Student or officer grading view
- /study — Study tools hub
- /study/pis — PI browser
- /study/flashcards — Flashcard sets
- /study/vocab — Vocab cards
- /study/theories — Theories & fallacies
- /study/visuals — Visual reference library
- /notes — Personal notes
- /leaderboard — Chapter leaderboard
- /profile — User profile & settings
- /admin — Officer/Advisor admin panel

7.2 Navigation

- Persistent top navbar: logo, main nav links, user avatar + dropdown
- Mobile-responsive hamburger menu
- Breadcrumbs on nested pages (e.g., Tests > Custom Test > Review)
- Quick-access sidebar on dashboard: recent tests, flagged PIs, unreviewed submissions

8. Database Schema (Supabase)

All tables live in Supabase (PostgreSQL). Row-Level Security (RLS) policies enforce role-based access.

Core Tables

users

- id (uuid, PK), first_name, last_name, email, password_hash
- role (student | officer | advisor)
- grade_level, chapter_id (FK), last_login, created_at
- total_points, avatar_url, is_public_on_leaderboard

chapters

- id, school_name, chapter_name, advisor_account_id (FK), state

events

- id, event_name, cluster, instructional_area, description
- prep_time_min, pres_time_min, allowed_material, rubric_total_points

scenarios

- id, event_id (FK), year, level (district | state | icdc)
- scenario_title, situation_description, student_instructions, judge_instructions
- pi_codes (array), source_url

performance_indicators

- id, pi_code, performance_indicator, performance_element
- instructional_area, cluster, tier

Test Tables

question, s

- question_text, choice_a/b/c/d, correct_choice
- rationale, pi_code (FK), cluster, instructional_area, level, difficulty, source

test_sessions

- id, user_id (FK), session_type (full | custom | pi_targeted)
- started_at, completed_at, score, total_questions, config (JSON)

test_answers

- id, session_id (FK), question_id (FK), chosen_answer, is_correct, time_spent_seconds

test_notes

- id, session_id (FK), user_id (FK), question_id (FK), note_text, created_at

Roleplay Tables

submissions

- id, user_id (FK), scenario_id (FK), event_id (FK)
- video_url, transcript_text, duration_seconds, attempt_number
- status (submitted | under_review | reviewed), submitted_at, reviewed_at

submission_comments

- id, submission_id (FK), author_id (FK)
- timestamp_seconds, comment_text, pi_code (optional), created_at

rubric_scores

- id, submission_id (FK), scorer_id (FK), event_id (FK)
- pi_scores (JSON: {pi_code: score}), twenty_first_century_score
- total_score, overall_feedback, scored_at

Study / Content Tables

vocab_terms

- id, term, definition, cluster, example_usage, source

flashcard_sets

- id, title, type (pi | vocab | custom), cluster, created_by (FK)

flashcards

- id, set_id (FK), front_text, back_text, pi_code (optional)

user_flashcard_progress

- id, user_id (FK), flashcard_id (FK), status (learning | know_it), last_seen

theories

- id, theory_name, category (motivational | psychological | fallacy)
- explanation, example_scenario, cluster_relevance

visuals

- id, title, description, cluster, image_url, source_type (chart | matrix | framework)

notes

- id, user_id (FK), tab_name, content (rich text/JSON), updated_at

Tracking & Gamification Tables

pi_performance

- id, user_id (FK), pi_code (FK), source (test | roleplay)
- total_attempts, correct_count, last_updated
- (Aggregated view: weak_pis view that surfaces PIs below 70% accuracy)

user_points_log

- id, user_id (FK), points_earned, action_type, reference_id, earned_at

announcements

- id, author_id (FK), message, visible_to (all | officers | students), created_at, expires_at
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9. AI Features (Future Implementation)

All AI features use the **Anthropic Claude API**. These are post-MVP but should be architected for in the database and UI from the start.

9.1 AI PI Coach

- User selects a PI and asks the AI to explain it
- AI provides: plain-English definition, real-world business example, common exam traps
- AI gives a short quiz to test comprehension, then logs result to pi_performance

9.2 AI Roleplay Scenario Generator

- Input: user's weak PI list (from their performance data) or any user-chosen PI
- Output: a custom roleplay scenario specifically targeting those PIs
- Scenario includes: situation description, student instructions, judge instructions
- Can be saved to the user's personal scenario library

9.3 AI Test Question Generator

- For PIs with few existing questions in the bank, AI generates new MCQ questions
- Questions flagged as AI-generated in the DB
- Officers can approve/reject AI questions for the main bank

9.4 AI Exam Specifications Coach

- Covers district-specific instructional areas, state exam specs, and ICDC specifications
- AI explains what is and isn't tested at each level
- Personalized study plan generator: given a competition level and date, creates a week-by-week study schedule

9.5 AI Flashcard Explainer

- User flips a flashcard and can ask 'Explain this more'
 - AI gives a deeper explanation, analogies, and a quick check question
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10. MVP Launch Checklist

Phase 1 — Foundation

1. Supabase project setup + all core tables created
2. User auth: sign up, login, logout, password reset
3. Role system: student / officer / advisor with RLS policies
4. Basic nav + routing across all main pages

Phase 2 — Tests

5. Seed question bank (manual + scraped)
6. Full test mode + immediate feedback mode
7. Calculator + notepad in-test
8. Post-test review with rationales
9. PI tracking from test answers

Phase 3 — Roleplays

10. Seed scenario library
11. Submission form with Google Drive video embed
12. Officer grading interface with timestamped comments
13. Rubric scoring form + student feedback view

Phase 4 — Study Tools

14. PI browser with filters
15. Flashcard sets (PI + Vocab) with learn/test/match modes
16. Theories & fallacies section with MCQ drills
17. Personal notes with rich text + auto-save

Phase 5 — Engagement & Polish

18. Points system wired to all actions
19. Chapter leaderboard (weekly + all-time)
20. Personal dashboard with PI heatmap + progress stats
21. Admin panel for officers/advisors

22. Mobile responsiveness pass

11. Open Questions & Notes

- Scraping legality: confirm terms of service on deca.org and decaplus.org before automated scraping; may need manual seeding for initial launch
- Video hosting: Google Drive embed is the MVP approach; evaluate Loom or direct upload (Supabase Storage) for v2
- Grader compensation: define officer grading expectations — is there a turnaround time SLA?
- PI sheet source of truth: confirm which PDF/version of the PI sheet to use as the canonical source for seeding the database
- Score normalization: DECA scaled scores are not publicly documented; consider tracking raw % only for MVP
- COPPA / FERPA: chapter members may be under 18; ensure no PII is exposed on the public leaderboard without opt-in